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Log Trailer Load Adjustment Assembly

Abstract

A log trailer load adjustment assembly includes a bolster and a plurality of links. The bolster has a crossbar. The plurality of links is at a first end pivotably coupled to a trailer frame and at a second end pivotably coupled to the crossbar such that the bolster can be moved forward or rearward.

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Background/Summary

FIELD OF THE INVENTION

[0001] The present invention relates generally to trailer rigs for hauling logs. Specifically, the present invention is a log trailer load adjustment assembly. The present invention is specifically designed for hauling logs. However, the present invention is not limited to this option, and it may further be adapted for carrying poles, beams and other lengthy items.

BACKGROUND OF THE INVENTION

[0002] A log trailer is a specialized type of plantation wagon designed specifically for transporting logs and timber on plantation estates. These trailers are essential in timber harvesting operations, allowing workers to efficiently move logs from the forest to processing areas or loading docks. Typically constructed with a sturdy flatbed and reinforced frame, log trailers can carry heavy loads of logs over rough terrain commonly found in wooded areas. They are often equipped with features such as adjustable stakes or bunks to secure the logs during transport, as well as heavy-duty wheels and axles to withstand the weight and demands of the job.

[0003] Log trailers used to be pulled by draft animals such as horses or oxen, although in modern times, they are typically pulled by tractors or other motorized vehicles. They play a crucial role in the timber industry, facilitating the efficient harvesting, transportation, and processing of logs for various purposes including lumber production, paper manufacturing, and fuelwood production.

[0004] However, the existing log trailers usually are unable to secure logs of varying lengths. It is an objective of the present invention to provide a log trailer load adjustment assembly that can adjust to accommodate logs of varying lengths.

SUMMARY OF THE INVENTION

[0005] The present invention discloses a log trailer load adjustment assembly. It comprises a bolster and a plurality of links. The bolster has a crossbar. The plurality of links is at a first end pivotably coupled to a trailer frame and at a second end pivotably coupled to the crossbar such that the bolster can be moved forward or rearward.

[0006] In one embodiment, the bolster further comprises a pair of bolter uprights extending from opposing ends of the crossbar.

[0007] In one embodiment, each of the plurality of links comprises an upper hole at the first end and a lower hole at the second end. The bolster comprises a plurality of upper link bases. The trailer frame comprises a plurality of lower link bases. The plurality of links is at the first end pivotably coupled to the plurality of upper link bases via a plurality of upper link pins traveling through the upper hole and at a second end pivotably coupled to the plurality of lower link bases via a plurality of lower link pins traveling through the lower hole.

[0008] In one embodiment, the plurality of links comprises a right set of links and a left set of links. The plurality of upper link bases comprises a right upper link base and a left upper link base, and the plurality of lower link bases comprises a right lower link base and a left lower link base.

[0009] In one embodiment, the right set of links and the left set of links each comprise two parallel links.

[0010] In one embodiment, the trailer frame comprises a forward horizontal section, a rearward horizontal section, and a transition section between the forward horizontal section and the rearward horizontal section. The plurality of lower link bases is located at the transition section.

[0011] In one embodiment, the forward horizontal section is higher than the rearward horizontal section, and the rearward horizontal section comprises at least one bunk having a plurality of bunk uprights. When moved forward, the bolster rests on the forward horizontal section, and when moved rearward, the bolster abuts against the plurality of bunk uprights.

[0012] In one embodiment, The plurality of bunk uprights each comprises a support block that is adapted to support the bolster when the bolster is moved rearward.

[0013] In one embodiment, the present invention further comprises a locking mechanism. The locking mechanism comprises a lock frame having a distal end and a proximal end, a hook portion beneath the lock frame and adjacent to the proximal end of the lock frame, and a pivotal connecting portion arranged between the hook portion and the lock frame and connected to the crossbar of the bolster via a plurality of lock mounting plates such that the locking mechanism pivots about a first axis. When the distal end of the lock frame is lowered, the hook portion engages a cross-brace of the trailer frame, and when the distal end of the lock frame is raised, the hook portion disengages from the cross-brace.

[0014] In one embodiment, the lock mechanism further comprises a lever having a distal end and a proximal end. The lever is pivotable about a second axis that travels through the proximal end of the lock frame. The proximal end of the lever comprises a weight that rests against the plurality of lock mounting plates and is received in a space between the crossbar of the bolster and the proximal end of the lock frame to prevent the lock frame from being lifted.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0015] The accompanying drawings, which are included to provide a further understanding of the invention are incorporated in and constitute a part of this specification, illustrate an embodiment of the invention and together with the description serve to explain the principles of the invention. They are meant to be exemplary illustrations provided to enable persons skilled in the art to practice the disclosure and are not intended to limit the scope of the present invention. That is, the dimensions of the components of the present invention, independently and in relation to each other can be different. It should be noted that the drawings are schematic and not necessarily drawn to scale. Some drawings are enlarged or reduced to improve drawing legibility.

[0016] FIG. **1** depicts a perspective view of the present invention, wherein the bolster rests on the forward horizontal section.

[0017] FIG. **2** depicts another perspective view of the present invention, wherein the bolster rests on the forward horizontal section.

[0018] FIG. **3** depicts yet another perspective view of the present invention, wherein the bolster rests on the forward horizontal section.

[0019] FIG. **4** depicts a right-side view of the present invention, wherein the bolster rests on the forward horizontal section.

[0020] FIG. **5** depicts a perspective view of the present invention, wherein the bolster rests on the forward horizontal section and the distal end of the lever is pressed down.

[0021] FIG. **6** depicts a right-side view of the present invention, wherein the bolster rests on the forward horizontal section and the distal end of the lever is pressed down.

[0022] FIG. **7** depicts a perspective view of the present invention, wherein the bolster rests on the forward horizontal section, the distal end of the lever is pressed down, and the distal end of the lock frame is raised.

[0023] FIG. **8** depicts a right-side view of the present invention, wherein the bolster rests on the forward horizontal section, the distal end of the lever is pressed down, and the distal end of the lock frame is raised.

[0024] FIG. **9** depicts a perspective view of the present invention, wherein the bolster is being moved rearward.

[0025] FIG. **10** depicts a right-side view of the present invention, wherein the bolster is being moved rearward.

[0026] FIG. **11** depicts a perspective view of the present invention, wherein the bolster is moved rearward to abut against the bunk, the distal end of the lever is pressed down, and the distal end of the lock frame is raised.

[0027] FIG. **12** depicts a right-side view of the present invention, wherein the bolster is moved rearward to abut against the bunk, the distal end of the lever is pressed down, and the distal end of the lock frame is raised.

[0028] FIG. **13** depicts a perspective view of the present invention, wherein the bolster is moved rearward to abut against the bunk, and the lever and the lock frame are returned to their neutral position due to gravity.

[0029] FIG. **14** depicts a right-side view of the present invention, wherein the bolster is moved rearward to abut against the bunk, and the lever and the lock frame are returned to their neutral position due to gravity.

[0030] FIG. **15** depicts a perspective view of the link of the present invention.

[0031] FIG. **16** depicts a perspective view of the lock frame of the present invention.

[0032] FIG. **17** depicts a perspective view of the locking mechanism of the present invention.

DETAIL DESCRIPTIONS OF THE INVENTION

[0033] As a preliminary matter, it will readily be understood by one having ordinary skill in the relevant art that the present disclosure has broad utility and application. As should be understood, any embodiment may incorporate only one or a plurality of the above-disclosed aspects of the disclosure and may further incorporate only one or a plurality of the above-disclosed features. Furthermore, any embodiment discussed and identified as being "preferred" is considered to be part of a best mode contemplated for carrying out the embodiments of the present disclosure. Other embodiments also may be discussed for additional illustrative purposes in providing a full and enabling disclosure. Moreover, many embodiments, such as adaptations, variations, modifications, and equivalent arrangements, will be implicitly disclosed by the embodiments described herein and fall within the scope of the present disclosure.

[0034] Accordingly, while embodiments are described herein in detail in relation to one or more embodiments, it is to be understood that this disclosure is illustrative and exemplary of the present disclosure and is made merely for the purposes of providing a full and enabling disclosure. The detailed disclosure herein of one or more embodiments is not intended, nor is to be construed, to limit the scope of patent protection afforded in any claim of a patent issuing here from, which scope is to be defined by the claims and the equivalents thereof. It is not intended that the scope of patent protection be defined by reading into any claim a limitation found herein that does not explicitly appear in the claim itself. Accordingly, it is intended that the scope of patent protection is to be defined by the issued claim(s) rather than the description set forth herein.

[0035] Additionally, it is important to note that each term used herein refers to that which an ordinary artisan would understand such term to mean based on the contextual use of such term herein. When not explicitly defined herein, to the extent that the meaning of a term used herein—as understood by the ordinary artisan based on the contextual use of such term—differs in any way from any particular dictionary definition of such term, it is intended that the meaning of the term as understood by the ordinary artisan should prevail.

[0036] Furthermore, it is important to note that, as used herein, "a" and "an" each generally denotes "at least one," but does not exclude a plurality unless the contextual use dictates otherwise. When used herein to join a list of items, "or" denotes "at least one of the items," but does not exclude a plurality of items of the list. Finally, when used herein to join a list of items, "and" denotes "all of the items of the list."

[0037] The following detailed description refers to the accompanying drawings. Wherever possible, the same reference numbers are used in the drawings and the following description to refer to the same or similar elements. While many embodiments of the disclosure may be described, modifications, adaptations, and other implementations are possible. For example, substitutions, additions, or modifications may be made to the elements illustrated in the drawings, and the methods described herein may be modified by substituting, reordering, or adding stages to the disclosed methods. Accordingly, the following detailed description does not limit the disclosure. Instead, the proper scope of the disclosure is defined by the appended claims. The present disclosure contains headers. It should be understood that these headers are used as references and are not to be construed as limiting upon the subject matter disclosed under the header.

[0038] Other technical advantages may become readily apparent to one of ordinary skill in the art after review of the following figures and description. It should be understood at the outset that, although exemplary embodiments are illustrated in the figures and described below, the principles of the present disclosure may be implemented using any number of techniques, whether currently known or not. The present disclosure should in no way be limited to the exemplary implementations and techniques illustrated in the drawings and described below.

[0039] Unless otherwise indicated, the drawings are intended to be read together with the specification and are to be considered a portion of the entire written description of this invention. As used in the following description, the terms "horizontal", "vertical", "left", "right", "up", "down" and the like, as well as adjectival and adverbial derivatives thereof (e.g., "horizontally", "rightwardly", "upwardly", "radially", etc.), simply refer to the orientation of the illustrated structure as the particular drawing figure faces the reader. Similarly, the terms "inwardly," "outwardly" and "radially" generally refer to the orientation of a surface relative to its axis of elongation, or axis of rotation, as appropriate. As used herein, the term "proximate" refers to positions that are situated close/near in relationship to a structure. As used in the following description, the term "distal" refers to positions that are situated away from positions.

[0040] The present disclosure includes many aspects and features. Moreover, while many aspects and features relate to, and are described in the context of log trailer load adjustment assemblies, embodiments of the present disclosure are not limited to use only in this context.

[0041] The present invention is a log trailer load adjustment assembly that is specifically designed log trailers. The present invention allows the existing log trailers to be retrofitted to accommodate logs of varying lengths. It is an aim of the present invention to provide users with a device that locks into place along the trailer frame to create a fixed bolster as needed. It is another aim of the present invention to provide a device that can move back and forth to allow for the difference of point-of-balance when loading logs of varying lengths.

[0042] Referring now to the figures of the present disclosure. The log trailer load adjustment assembly of the present invention comprises a bolster **100** having a crossbar **110**, and a plurality of links **120**.

[0043] The bolster **100** is adapted to coupled to a trailer frame **200**. It should be noted that the bolster **100** can be of any shape, size, material, features, type or kind, orientation, location, quantity, components, and arrangements of components that would allow the present invention to fulfill the objectives and intents of the present invention. The crossbar **110** is supported on the trailer frame **200**. Typically, the crossbar **110** is wider than the trailer frame **200** and extends transversely to the trailer frame **200**. In one embodiment, the bolster **100** further comprises a pair of bolter uprights **112** extending from opposing ends of the crossbar **110**. The crossbar **110**, together with the bolster uprights **112**, forms a substantially U-shaped structure to receive and support a payload of logs.

[0044] The plurality of links **120** is configured to couple the bolster **100** to the trailer frame **200**. It should be noted that the plurality of links **120** can be of any shape, size, material, features, type or kind, orientation, location, quantity, components, and arrangements of components that would allow the present invention to fulfill the objectives and intents of the present invention. In one embodiment, each of the plurality of links **120** is a rectangular cross-sectional tubing. However, it is contemplated that that the cross section may take on other shapes, including but not limited to square, circular, triangular, or other regular or irregular polygonal forms. The plurality of links **120** is designed with a metal material that provides structural strength. The plurality of links **120** is at a first end **121** pivotably coupled to the trailer frame **200** and at a second end **122** pivotably coupled to the crossbar **110** such that the bolster **100** can be moved forward or rearward. Therefore, a user can adjust the positioning of the bolster **100** along the trailer frame **200** as needed to accommodate logs of various lengths.

[0045] In one embedment, each of the plurality of links **120** comprises an upper hole **123** at the first end **121** and a lower hole **124** at the second end **122**. The bolster **100** comprises a plurality of upper link bases **126**, and the trailer frame **200** comprises a plurality of lower link bases **127**. The upper link base **126** may be a rectangular shaped bracket that is fixed to the crossbar **110**. The lower link base **127** may be designed with a similar shape and fixed to the outside of the trailer frame **200**. The plurality of links **120** is at the first end **121** pivotably coupled to the plurality of upper link bases **126** via a plurality of upper link pins **128** traveling through the upper hole **123** and at a second end **122** pivotably coupled to the plurality of lower link bases **127** via a plurality of lower link pins **129** traveling through the lower hole **124**. In one embodiment, the plurality of links **120** comprises a right set of links **120***a* and a left set of links **120***b*. The plurality of upper link bases **126** comprises a right upper link base **126***a* and a left upper link base **126***b*, and the plurality of lower link bases **127** comprises a right lower link base **127***a* and a left lower link base **127***b*. In a preferred embodiment, the right set of links **120***a* and the left set of links **120***b* each comprise two parallel links. The parallel links will remain parallel as they pivot about the lower link bases **127**. The right set of links **120***a* and the left set of links **120***b* pivot forward or rearward simultaneously, allowing the bolster **100** to move forward or rearward accordingly.

[0046] The trailer frame **200** comprises a forward horizontal section **210**, a rearward horizontal section **220**, and a transition section **230** between the forward horizontal section **210** and the rearward horizontal section **220**. The plurality of lower link bases **127** is located at the transition section **230**. In one embodiment, the forward horizontal section **210** is higher than the rearward horizontal section **220**. The transition section **230** may be an angled off-set section. The forward horizontal section **210** is higher than the rearward horizontal section **220**. The trailer frame **200** typically comprises two longitudinal side beams that lie parallel to one another and are interconnected together by a plurality of cross-braces. It is important to note that the figures depict only a portion of the trailer frame **200**, and the lengthwise dimension of the trailer frame **200** significantly exceeds its width.

[0047] In one embodiment, the rearward horizontal section **220** comprises at least one bunk **221** having a plurality of bunk uprights **222**. When moved forward, the bolster **100** rests on the forward horizontal section **210** as shown in FIGS. **1-4**, and when moved rearward, the bolster **100** abuts against the plurality of bunk uprights **222** as shown in FIGS. **11-14**. In a preferred embodiment, the plurality of bunk uprights **222** each comprises a support block **223** that is adapted to support the bolster **100** when the bolster **100** is moved rearward.

[0048] In a preferred embodiment, the present invention further comprises a locking mechanism **140**. The locking mechanism **140** comprises a lock frame **141**, a hook portion **145**, and a pivotal connecting portion **146**. The lock frame **141** has a distal end **142** and a proximal end **143**. The hook portion **145** is beneath the lock frame **141** and adjacent to the proximal end **143** of the lock frame **141**. The pivotal connecting portion **146** is arranged between the hook portion **145** and the lock frame **141** and connected to the crossbar **110** of the bolster **100** via a plurality of lock mounting plates **130** such that the locking mechanism pivots about a first axis **147**. The first axis **147** travels through the plurality of lock mounting plates **130** and runs parallel to the lengthwise dimension of the crossbar **110** of the bolster **100**. The distal end **142** of the lock frame **141** can be raised or lowered, for example using a log loader, to operate the locking mechanism. When the distal end **142** of the lock frame **141** is lowered, the hook portion **145** engages a cross-brace **212** of the trailer frame **200**, preventing the bolster **100** from moving, and when the distal end **142** of the lock frame **141** is raised, the hook portion **142** disengages from the cross-brace **212**, allowing the bolster **100** to move.

[0049] In one embodiment, the lock mechanism **140** further comprises a lever **150** having a distal end **152** and a proximal end **153**. The lever is pivotable about a second axis **157** that travels through the proximal end **143** of the lock frame **141**. The proximal end **153** of the lever **150** comprises a weight **155** that rests against the plurality of lock mounting plates **130** and is received in a space **156** between the crossbar **110** of the bolster **100** and the proximal end **143** of the lock frame **141** to prevent the lock frame **141** from being lifted. Due to gravity, the weight **155** creates a rotational

force or torque that tends to rotate the lever in a downward direction, positioning the weight **155** within the space **156** when no external forces are applied to the distal end **152** of the lever **150**. To unlock the locking mechanism, the user may use a log loader to first press down the distal end **152** of the lever **150**, lifting the weight **155** out of the space **156**. Next, the distal end **142** of the lock frame **141** may be raised by the log loader to disengage the hook portion **145** from the cross-brace **212**. The log loader may then pull the lock frame **141** rearward to move the bolster **100** in the same direction. When the bolster **100** is moved to the desired position, the log loader disconnects from the lock frame, allowing the distal end **142** of the lock frame **141** to lower under the force of gravity. Simultaneously, the weight **155** of the lever **150** returns to the space **156**, resting on the plurality of lock mounting plates **130**. In this manner, the locking mechanism automatically re-engages and locks into place.
[0050] The present invention may be made of any structural rigid material such as cast aluminum, steel or any other suitable materials or using any suitable combination thereof. It should be understood that the components of the present invention may or may not be made of the same material. Further, it is envisioned that the sizes of the components forming the present invention can vary based on design requirements.
[0051] Although the disclosure has been explained in relation to its preferred embodiment, it is to be understood that many other possible modifications and variations can be made without departing from the spirit and scope of the disclosure.

Claims

1. A log trailer load adjustment assembly comprising: a bolster having a crossbar; a plurality of links; the plurality of links being at a first end pivotably coupled to a trailer frame and at a second end pivotably coupled to the crossbar such that the bolster can be moved forward or rearward.
2. The log trailer load adjustment assembly as claimed in claim 1, wherein the bolster further comprises a pair of bolter uprights extending from opposing ends of the crossbar.
3. The log trailer load adjustment assembly as claimed in claim 1, wherein each of the plurality of links comprises an upper hole at the first end and a lower hole at the second end, the bolster comprises a plurality of upper link bases, and the trailer frame comprises a plurality of lower link bases, the plurality of links being at the first end pivotably coupled to the plurality of upper link bases via a plurality of upper link pins traveling through the upper hole and at a second end pivotably coupled to the plurality of lower link bases via a plurality of lower link pins traveling through the lower hole.
4. The log trailer load adjustment assembly as claimed in claim 3, wherein the plurality of links comprises a right set of links and a left set of links, wherein the plurality of upper link bases comprises a right upper link base and a left upper link base, and the plurality of lower link bases comprises a right lower link base and a left lower link base.
5. The log trailer load adjustment assembly as claimed in claim 4, wherein the right set of links and the left set of links each comprise two parallel links.
6. The log trailer load adjustment assembly as claimed in claim 3, wherein the trailer frame comprises a forward horizontal section, a rearward horizontal section, and a transition section between the forward horizontal section and the rearward horizontal section, the plurality of lower link bases being located at the transition section.
7. The log trailer load adjustment assembly as claimed in claim 6, wherein the forward horizontal section is higher than the rearward horizontal section, and the rearward horizontal section comprises at least one bunk having a plurality of bunk uprights, wherein when moved forward, the bolster rests on the forward horizontal section, and when moved rearward, the bolster abuts against the plurality of bunk uprights.
8. The log trailer load adjustment assembly as claimed in claim 7, wherein The plurality of bunk

uprights each comprises a support block that is adapted to support the bolster when the bolster is moved rearward.

9. The log trailer load adjustment assembly as claimed in claim 8, further comprising a locking mechanism that comprises: a lock frame having a distal end and a proximal end; a hook portion beneath the lock frame and adjacent to the proximal end of the lock frame; a pivotal connecting portion arranged between the hook portion and the lock frame and connected to the crossbar of the bolster via a plurality of lock mounting plates such that the locking mechanism pivots about a first axis; wherein when the distal end of the lock frame is lowered, the hook portion engages a cross-brace of the trailer frame, and when the distal end of the lock frame is raised, the hook portion disengages from the cross-brace.

10. The log trailer load adjustment assembly as claimed in claim 9, wherein the lock mechanism further comprises a lever having a distal end and a proximal end, the lever being pivotable about a second axis that travels through the proximal end of the lock frame, the proximal end of the lever comprising a weight that rests against the plurality of lock mounting plates and is received in a space between the crossbar of the bolster and the proximal end of the lock frame to prevent the lock frame from being lifted.